Vibration Monitoring and GPS Alerts for Track Defects

Abstract.

We study whether accelerometers on an inspection trolley can flag track defects and report their location in real time. The system flagged 87% of known defects with a median alert delay of 9 seconds.

Method.

Two MEMS accelerometers sampling at 2 kHz are mounted on the trolley axle. An anomaly score compares each 0.5 s window with a rolling baseline, and windows above a threshold trigger an alert. A GSM module sends each alert as a text message with the GPS position.

Data.

Trials ran over a 6 km test track with 46 known defects, including cracks, loose fastenings and dipped joints.

Note.

A fictional sample paper by P. Novak and R. Das (2025), written to test MAIROS. It is not published research: its data and results are invented.

Results.

The system flagged 40 of the 46 defects (87%) and raised about one false alarm every 3 km. The median delay from passing a defect to the alert arriving was 9 seconds. Weak GSM coverage delayed 12% of alerts by more than one minute.

Limitations.

A limitation of this study is that it used a single 6 km test track, so the false-alarm rate may not hold on busy mainline routes. Vibration alone could not tell a crack from a loose fastening.

Open questions.

An open problem is how to fuse vibration alerts with visual confirmation to cut false alarms. Future work should also test alert delivery where there is no GSM signal.

Conclusion.

Low-cost accelerometers can flag defects and send located alerts, but they need a second sensor to classify what they found.